

Measuring the pH and turbidity levels of Lactobacillus acidophilus and Escherichia coli in the Presence of Microplastics

Question
Do microplastics and nanoplastics cause distinct metabolic changes in Gram-positive and Gram-negative gut bacteria, and can these changes model disease-associated conditions observed in diseases, such as depression and colorectal cancer?

Hypothesis
Exposure to polystyrene and polyethylene microplastics will disrupt the metabolic activity of gut bacteria, lowering culture pH and decreasing bacterial survival, with Gram-positive and Gram-negative species responding differently based on their structures, collectively producing conditions of dysbiosis.

Introduction
Microplastics, tiny plastic particles under 5 mm, have exploded as environmental pollutants amid booming production, from 2 million tonnes in 1950 to ~400 million tons yearly today. About 8-12 million tons enter oceans annually (UNEP, 2021), breaking into secondary microplastics via weathering/UV or manufactured as primary ones like cosmetic microbeads. They’ve invaded humans, detected in blood, lungs, placentas, and feces, with an average ingestion of 39,000-52,000 particles yearly via food/water. In addition, the average human ingests around 90,000 from bottled water alone (Cox et al., 2019; Qian et al., 2019). Most exposure hits the gut, home to around 100 trillion bacteria across over 1,000 species that digest food, produce vitamins, regulate immunity, and ferment fiber into short-chain fatty acids (SCFAs: acetate, propionate, butyrate). Dominated by Firmicutes/Bacteroidetes (80-90%), key players like <i>Lactobacillus</i>, <i>Bifidobacterium</i> (lactic acid for pH balance), <i>Bacteroides</i> (carb breakdown), and <i>Faecalibacterium</i> (anti-inflammatory butyrate) keep colon pH at 5.5-6.5. This acidity favors beneficial bacteria while curbing pathogens; SCFAs also fuel cells, strengthen barriers, control blood sugar/cholesterol/appetite, and reduce inflammation. Dysbiosis disrupts this, linking to IBD, obesity, diabetes, colorectal cancer, and, via the gut-brain axis, even depression/anxiety. TSimilarly, people with depression tend to have lower microbial diversity, fewer beneficial butyrate-producing bacteria, and higher levels of inflammation, suggesting the microbiome plays a role in mental health. Animal studies suggest that ingesting microplastics can disrupt the gut microbiome, causing dysbiosis, or an imbalance in gut bacteria, marked by reduced diversity and fewer beneficial microbes. Researchers believe this occurs because microplastics can damage gut cells and the protective mucus lining, provide surfaces for harmful biofilms to form, and release chemical additives that affect microbial activity. A key October 2025 study in Environmental Science & Technology exposed human fecal bacteria to common microplastics and found that the cultures became more acidic (lower pH) and showed shifts in microbial composition and metabolism similar to patterns seen in microbiomes associated with depression and colorectal cancer. The effects also vary by plastic type: polyethylene has been linked to metabolic changes in mice, while polystyrene appears to cause greater intestinal barrier damage and inflammation Healthy colon pH (5.5-6.5) signals balance—elevated in IBD/cancer—making it a perfect gauge for microplastic disruption of SCFA production and bacterial metabolism. This project tests PS fragments (from weighing boats) and PE microbeads on Gram-positive (<i>Lactobacillus</i>, <i>Bifidobacterium</i>; thick walls) vs. Gram-negative (<i>E. coli</i>, <i>Bacteroides</i>; thin walls) bacteria in tap/distilled water, probing contamination differences.

Methodology
To begin the experiment, prepare the laboratory workspace by organizing all materials required for microplastic preparation, bacterial culturing, and metabolic testing. Read the Standard Operating Procedure and put on proper PPE, including safety goggles, a buttoned lab coat, and double gloves to ensure full protection during handling of microplastics and live cultures. Collect the three water types needed for the experiment—tap water, bottled water, and boiled or filtered water—and prepare microplastic solutions at low, medium, and high concentrations using polyethylene or polystyrene beads sourced from an approved educational supplier. Confirm microplastic presence and particle size with sieving, density separation, and fluorescence-based Nile red staining under a dissecting microscope. After the exposure conditions are prepared, obtain the bacterial cultures required for the study: <i>Lactobacillus acidophilus</i> (Gram-positive), <i>Escherichia coli</i> K-12 (Gram-negative), and a probiotic composite gut culture, maintaining all strains under Biosafety Level 1 guidelines. Grow each culture in its appropriate broth (MRS for <i>Lactobacillus</i>, LB for <i>E. coli</i>, and BHI for the composite microbiome), keeping incubators at the correct temperatures to ensure uniform growth across samples. Once the bacteria reach the appropriate phase, transfer 12 mL aliquots of each culture into sterile tubes and expose them to the corresponding control, low, medium, or high microplastic concentrations for the full duration of the trial, ensuring that each bacterial group has a minimum of three independent replicates per condition. After exposure, measure the initial and final pH of each sample using calibrated pH instruments, and collect supernatants for metabolite analysis using colorimetric or spectrophotometric assays to quantify short-chain fatty acids and related fermentation products. Record changes in turbidity, pH, metabolite concentrations, and any visible aggregation around microplastic particles. Throughout the procedure, maintain strict contamination control by working on disinfected surfaces, properly sterilizing pipettes and loops, and ensuring that all microplastic-containing liquids and solids are discarded in designated sealed waste containers rather than down laboratory drains. After all data are collected, deactivate all cultures with chemical disinfectants or heat treatment and clean the workspace thoroughly.

Data/Results

Materials
Probiotic capsules
E. coli K-12
Bacillus subtilis
MRS or BHI broth (for probiotic mix)
Nutrient broth
Polyethylene (PE) microspheres
Polystyrene (PS) microspheres
Polypropylene (PP) fragments or beads
Petri dishes
Inoculating loops or sterile swabs
Incubator (37°C)
Spectrophotometer (OD ₆₀₀) and cuvettes
pH meter or pH strips
Compound light microscope
Centrifuge and microcentrifuge tubes

Conclusion

Acknowledgements
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